

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Sejal Singh
- b. Build and deploy a **single AI agent** using **ADK and Gemini** that is hosted on **Cloud Run** and performs **one clearly defined task**. The agent must be callable via an HTTP endpoint and return a valid response for a given input.

Brief about the idea

- This project is a simple AI-based **Text Summarization Agent**.
- It takes a long paragraph or text as input and generates a **short and meaningful summary**.
- The agent is built using **Google ADK** and powered by the **Gemini AI model**.
- It is deployed on **Google Cloud Run**, so it can be accessed anytime through an **HTTP API link**.
- The main aim is to make reading easier by converting lengthy content into quick summaries.

Meeting the Build Criteria

- I selected a **Text Summarization AI Agent** because it matches the “one clearly defined task” requirement.
- I used **Google ADK** to build the agent structure and define the agent instructions properly.
- I integrated the agent with the **Gemini model** for generating summaries from user input text.
- I deployed the agent on **Google Cloud Run**, so it works as a hosted cloud service.
- I made sure the agent is **callable through an HTTP endpoint**, where it accepts input and returns a valid summarized output.
- I kept the logic simple and focused mainly on **agent setup + deployment**, as required in the mini project criteria.

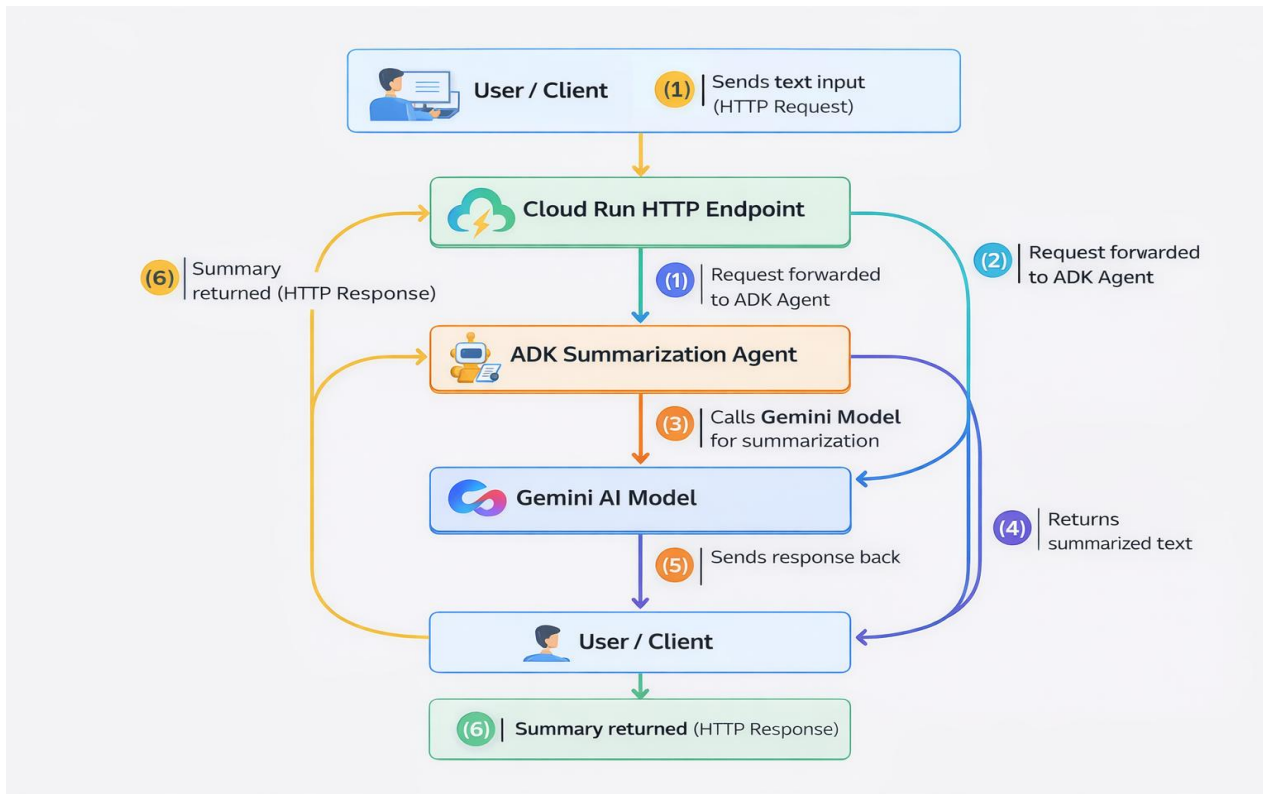
Opportunities

- How different is it from any of the other existing ideas?
 - How will it be able to solve the problem?
 - USP of the proposed solution
-
- Unlike normal summarizers, this solution is deployed as a **cloud-based AI agent** that can be accessed anytime using an API.
 - It helps users quickly understand long text by converting it into a **short and clear summary**, saving time and effort.
 - **Fast, simple, and accessible through an HTTP endpoint**
 - Can be integrated easily into any website/app
 - Powered by **Gemini**, so summaries are more accurate and meaningful

List of features offered by the solution

- Accepts long text as input
- Generates short and meaningful summary
- Uses **Gemini AI model** for accurate summarization
- Built using **Google ADK** (Agent Development Kit)
- Hosted on **Google Cloud Run** (cloud-based deployment)
- Accessible through an **HTTP API endpoint**
- Provides fast response in real-time
- Can be integrated with websites or applications easily

Process flow diagram or Use-case diagram



Technologies to be used in the solution

- **Google ADK (Agent Development Kit)** – for building the AI agent
- **Gemini Model (Vertex AI)** – for text summarization and AI inference
- **Google Cloud Run** – for hosting and deploying the agent as a web service
- **Python** – for coding the agent logic
- **Cloud Build** – for building and deploying the container automatically
- **Artifact Registry** – for storing Docker container images
- **HTTP REST API** – for sending input text and receiving summary output
- **Google Cloud Logging** – for monitoring and debugging the deployed service

Snapshots of the prototype

text-summarizer-16586071817.europe-west1.run.app/dev-ui/?app=text_summarizer_agent&session=9b4abfae-dafc-4908-8307-7a46b27534bb

Agent Development Kit

text_summarizer_ag...

Trace Events State Artifacts Sessions Eval

Invocations

Artificial Intelligence is changing the world rapidly. It is being u

Web development refers to building, creating, and maintaining

SESSION ID 9b4abfae-dafc-4908-8307-7a46b27534bb

Token Streaming | + New Session

Developing a web application involves more than coding; version control manages code changes, while deployment makes the application accessible to users, often integrated through automated workflows.

Version Control: Tracks and manages changes in the codebase, essential for teamwork.

Deployment: Publishes the web application so it's accessible online.

Automation: Modern workflows combine version control and deployment for efficiency and reliability. Ensures a smooth, organized, and maintainable development process.

Graphics

Graphical elements are one of the key feature of any webpage. They can be used to convey important points better than text does and beautify the webpage.

Canvas: The HTML "canvas" element is used to draw graphics via JavaScript.

SVG: SVG stands for Scalable Vector Graphics. It basically defines vector-based graphics in XML format.

Web development involves building and maintaining websites, encompassing design, programming, and database management. Websites function through browsers sending requests to web servers, which process data (often with databases) and return web pages. The field is divided into two main parts:

1. **Frontend Development:** Focuses on what users see and interact with, using HTML for structure, CSS for styling, and JavaScript for interactivity.
2. **Backend Development:** Handles server-side logic, data processing, and storage. It uses server-side programming languages (e.g., Node.js, PHP, Python, Java) and databases (SQL like MySQL/PostgreSQL, and NoSQL like MongoDB/Redis) to manage and store data.

Additionally, web development involves using APIs and data formats (like JSON/XML) for communication, version control and deployment for managing code and making applications accessible, and graphics (Canvas, SVG) for visual elements.

Type a Message...

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Google Cloud 

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Thank you

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